

Web Design and Development

Chapter Two

Fundamentals of Client-Side Web Technologies

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Sections

- **Part 1** : Hyper Text Mark-up Language (HTML), HTML5,
- **Part 2** : Cascading Style Sheets (CSS)
- **Part 3** : Introduction to JavaScript

Part 1 : Hyper Text Mark-up Language (HTML)

- What is HTML?
- Creating HTML File
- HTML Document Structure
- Basic HTML Tags
- Character Entities in HTML
- Creating Lists & Tables in HTML
- Creating Links
- Inserting an Image into a given page
- Creating Forms

What is HTML?

- Hyper Text Markup Language
- A markup language designed for the creation of web pages and other information viewable in a browser
- The basic language used to write web pages
- File extension: .htm, .html

Creating a HTML File

■ Creating a HTML File

1. Open Notepad or any other web editor
2. Click on File -> Save as...
3. In the File name pull-down box, type in `webpage.html`
4. Click on Save
5. Type in content for your file
6. Once you finished the content, click on File -> Save

HTML Document Structure

- `<html>`

- `<head>`

- `<title>` *Page Title Goes Here* `</title>`

- `</head>`

- `<body>`

- content goes here*

- `</body>`

- `</html>`

HTML Basics (1)

- `<html>`

```
<head>  
  <TITLE>A Simple HTML Example</TITLE>  
</head>
```

```
<body>  
  <H1>HTML is Easy To Learn</H1>  
  <P>Welcome to the world of HTML. This is the  
    first paragraph. While short it is still a  
    paragraph!  
  </P>  
  <P>And this is the second paragraph.</P>  
</body>  
</html>
```

HTML Basics (2)

▪ Markup Tags

- Most Tag usually goes with pair: **an open tag () and an end tag ()**
- Single tag: <hr>,

- Tags are NOT case sensitive

Effect	Code	Code Used	What It Does
Bold	B	Bold	Bold
Italic	I	<I>Italic</I>	<i>Italic</i>

HTML Basics (3)

▪ Markup Tags (Continued ...)

▪ HTML

This element tells your browser that the file contains HTML-coded information.

The file extension .html also indicates this an HTML document and must be used. (If you are restricted to 8.3 filenames (e.g., LeeHome.htm, use only .htm for your extension.)

▪ HEAD

The head element identifies the first part of your HTML-coded document that contains the title. The title is shown as part of your browser's window

HTML Basics (4)

- **Markup Tags (Continued...)**

- **TITLE**

The title element contains your document title and identifies its content in a global context. The title is typically displayed in the title bar at the top of the browser window, but not inside the window itself.

- **BODY**

The second--and largest--part of your HTML document is the body, which contains the content of your document (displayed within the text area of your browser window).

All other tags now come within the body tag.

HTML Basics (5)

- **Markup Tags (Continued ...)**

- **Headings**

HTML has six levels of headings, numbered 1 through 6, with 1 being the largest. Headings are typically displayed in larger and/or bolder fonts than normal body text. The first heading in each document should be tagged `<H1>`.

The syntax of the heading element is:

`<Hy>Text of heading </Hy>`

where *y* is a number between 1 and 6 specifying the level of the heading.

HTML Basics (6)

- Markup Tags (Continued ...)

- Headings

- The 6 heading commands.

<H1>This is Heading 1</H1>

<H2>This is Heading 2</H2>

<H3>This is Heading 3</H3>

<H4>This is Heading 4</H4>

<H5>This is Heading 5</H5>

<H6>This is Heading 6</H6>

HTML Basics (7)

- **Markup Tags (Continued ...)**

- **Paragraphs**

Unlike documents in most word processors, carriage returns in HTML files aren't significant.

In fact, any amount of *whitespace* – including spaces, linefeeds, and carriage returns -are automatically compressed into a single space when your HTML document is displayed in a browser.

A Web browser ignores this line break and starts a new paragraph only when it encounters another `<P>` tag

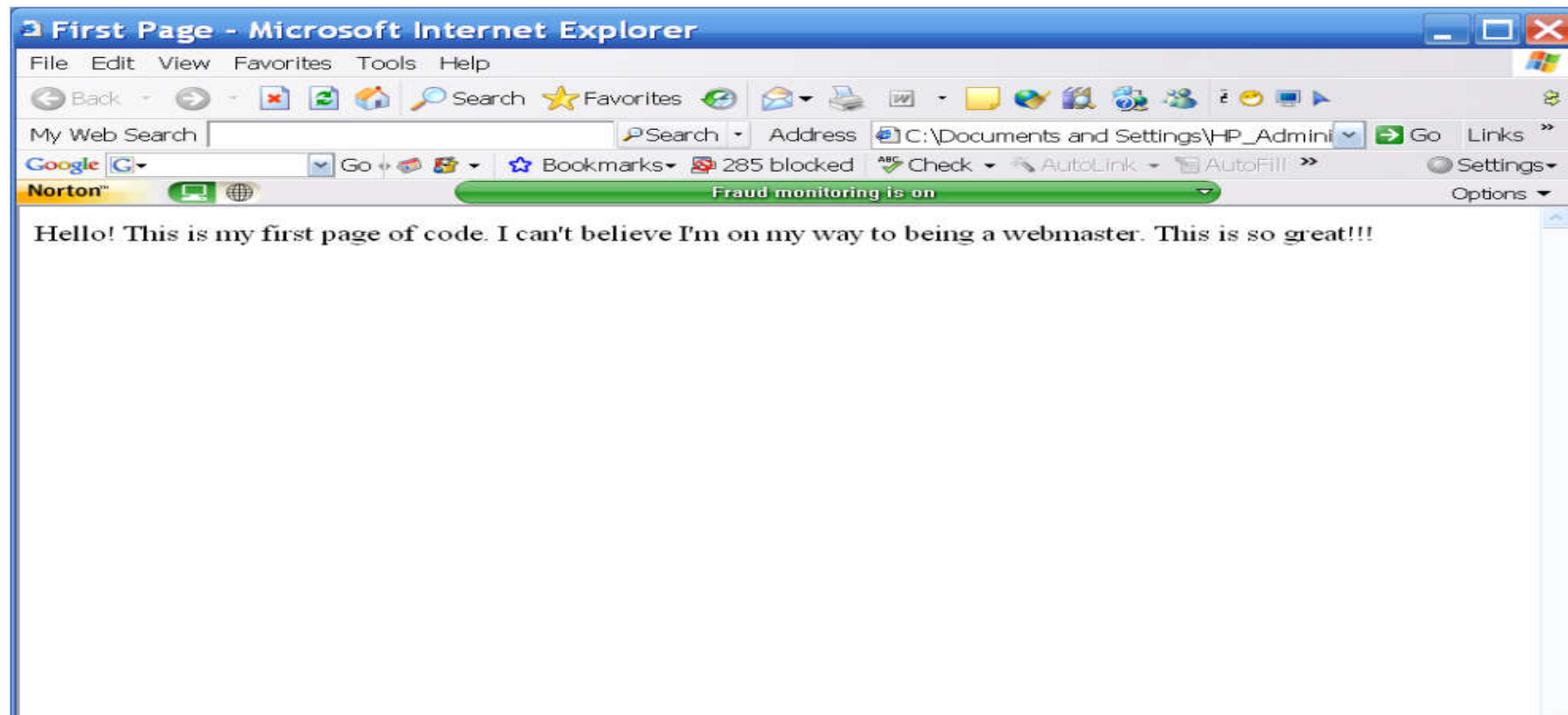
HTML Basics (8)

■ Simple page of Code

```
<html>  
  <head><title>First Page</title></head>  
  <body>  
    Hello! This is my first page of code. I can't  
    believe I'm on my way to being a webmaster.  
    This is so great!!!  
  </body>  
</html>
```

HTML Basics (9)

- What does that look like?



Commonly Used Character Entities

Result	Description	Entity Name
	Non-breaking space	
<	Less than	<
>	Greater than	>
&	Ampersand	&
“	Quotation mark	"
©	Copyright	©

Lists

- Unordered list

- Code:

- `<ul>`

- `<li>Coffee</li>`

- `<li>Milk</li>`

- `</ul>`

- Output:

- Coffee

- Milk

- Ordered list

- Code:

- `<ol>`

- `<li>Coffee</li>`

- `<li>Milk</li>`

- `</ol>`

- Output:

- 1. Coffee

- 2. Milk

Table

```
<table border="1">
<tr>
<th>Heading</th>
<th>Another Heading</th>
</tr>
<tr>
<td>row 1, cell 1</td>
<td>row 1, cell 2</td>
</tr>
<tr>
<td>row 2, cell 1</td>
<td></td>
</tr>
```

Heading	Another Heading
Row 1, cell 1	Row 1, cell 2
Row 2, cell 1	

Inserting an Image

- Place all images in the same directory/folder where your web pages are
- `<img src>` is a single tag
- ``
 - Output: 
- Turn an image into a hyperlink
 - `<a href="http://www.iusb.edu"></a>`
 - Output: 

Forms

- **Next Class**

Practice Time

- Create an practice.html page
- Practice by creating the html elements we have seen so far
- Ps : show your progress to your instructor

Forms - The <form> Element

- <Form> is HTML element used to collect data from users.
- <form> </form>
- Commonly used form elements includes:
 - Text fields
 - Radio buttons
 - Checkboxes
 - Text boxes
 - Submit buttons

Forms - The `<input>` Element

- The `<input>` element is the most important form element.
- The `<input>` element can be displayed in several ways, depending on the type attribute.
- Here are some examples:

Type	Description
<code><input type="text"></code>	Defines a one-line text input field
<code><input type="radio"></code>	Defines a radio button (for selecting one of many choices)
<code><input type="submit"></code>	Defines a submit button (for submitting the form)

Forms - `<input type="Text">` - Text Fields

- Used when you want the user to type letters, number, etc.
- `<form>`
First name: `<input type="text" name="firstname">`
`<br>`
Last name: `<input type="text" name="lastname">`
`</form>`

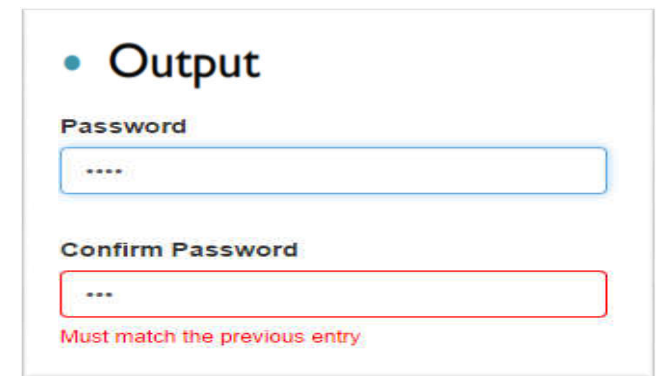
• Output

First name:

Last name:

Forms - `<input type="Text" >` - Text Fields

- Used when you want the user to type letters, number, and they are hidden.
- `<form>`
 Password: `<input type="password" name="password">`
 `<br>`
 Confirm password: `<input type="password" name="confirm">`
 `</form>`



The screenshot shows a web form titled "Output". It contains two password input fields. The first field is labeled "Password" and has a blue border. The second field is labeled "Confirm Password" and has a red border. Below the second field, there is a red error message that says "Must match the previous entry".

Forms - `<input type="radio">` - Radio Buttons

- Used when you want the user to select one of a limited number of choices.
- `<form>`
 `<input type="radio" name="sex" value="male"> Male`
 `<br>`
 `<input type="radio" name="sex" value="female"> Female`
`</form>`

• Output

- ☐ Male
- ☐ Female

Forms - `<input type="checkbox">` - Checkboxes

- Used when you want the user to select one of a limited number of choices.
- `<form>`
 `<input type="radio" name="sex" value="bike"> I have a bike`
 `<br>`
 `<input type="radio" name="sex" value="car"> I have a car`
`</form>`

• Output

- ☐ I have a bike
- ☐ I have a car

Forms - Text Boxes

- Used when you want to get input from user.

```
<form>
```

```
<p>Please provide your suggestion in the text box below:</p>
```

```
<textarea row="10"
```

```
cols="30">
```

```
</textarea>
```

```
</form>
```

• Output

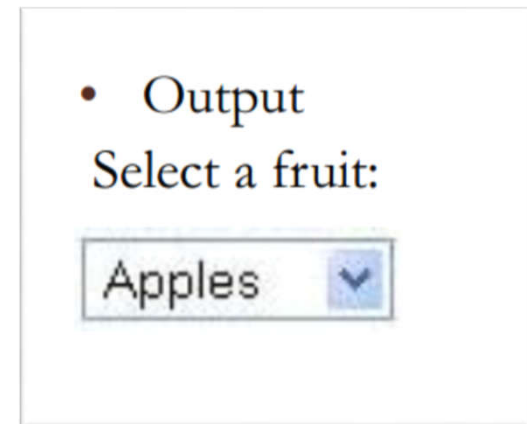
Please provide your
suggestion in the text box
below:

A rectangular text box with a thin black border, intended for user input.

Forms - Select/Drop down boxes

- Used when you want user to respond with one specific answer with choices you given.

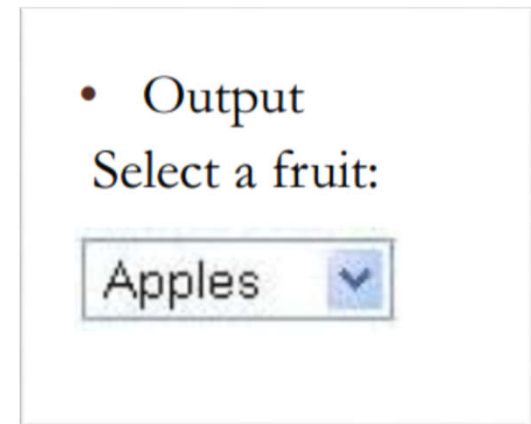
```
<p>Select a fruit:</p>  
<select name ="Fruit">  
    <option selected> Apples </option>  
    <option> Bananas </option>  
    < option > Oranges </option>  
</select>
```



Forms - Select/Drop down boxes

- Used when you want user to respond with one specific answer with choices you given.

```
<p>Select a fruit:</p>  
<select name ="Fruit">  
    <option selected> Apples </option>  
    <option> Bananas </option>  
    < option > Oranges </option>  
</select>
```



Forms - Submit buttons

- When user clicks on the “Submit” button, the content of the form is sent to another file.

```
<form name="input" action="html_form_acti on.asp" method="get">
```

```
Username: <input type="text" name="user">
```

```
<br>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

- Output

Username:

Submit

Forms - <Form> Tag attributes

- When user clicks on the “Submit” button, the content of the form is sent to another file.

```
<form name="input" action="html_form_action.asp" method="get">
```

```
Username: <input type="text" name="user">  
<br>
```

```
<input type="submit" value="Submit">  
</form>
```

• Output

Username:

HTML – Attributes (1)

- All HTML elements can have **attributes**
- Attributes provide **additional information** about an element
- Attributes are always specified in **the start tag**
- Attributes usually come in name/value pairs like: **name="value"**

HTML – Attributes (2)

- **The href Attribute :**

- HTML links are defined with the <a> tag. The link address is specified in the href attribute

```
<a href="https://www.w3schools.com">This is a link</a>
```

- **The alt Attribute :**

- The alt attribute specifies an alternative text to be used, when an image cannot be displayed.

```

```

End of Chapter Two

- **Your Questions are Very Welcome !!**